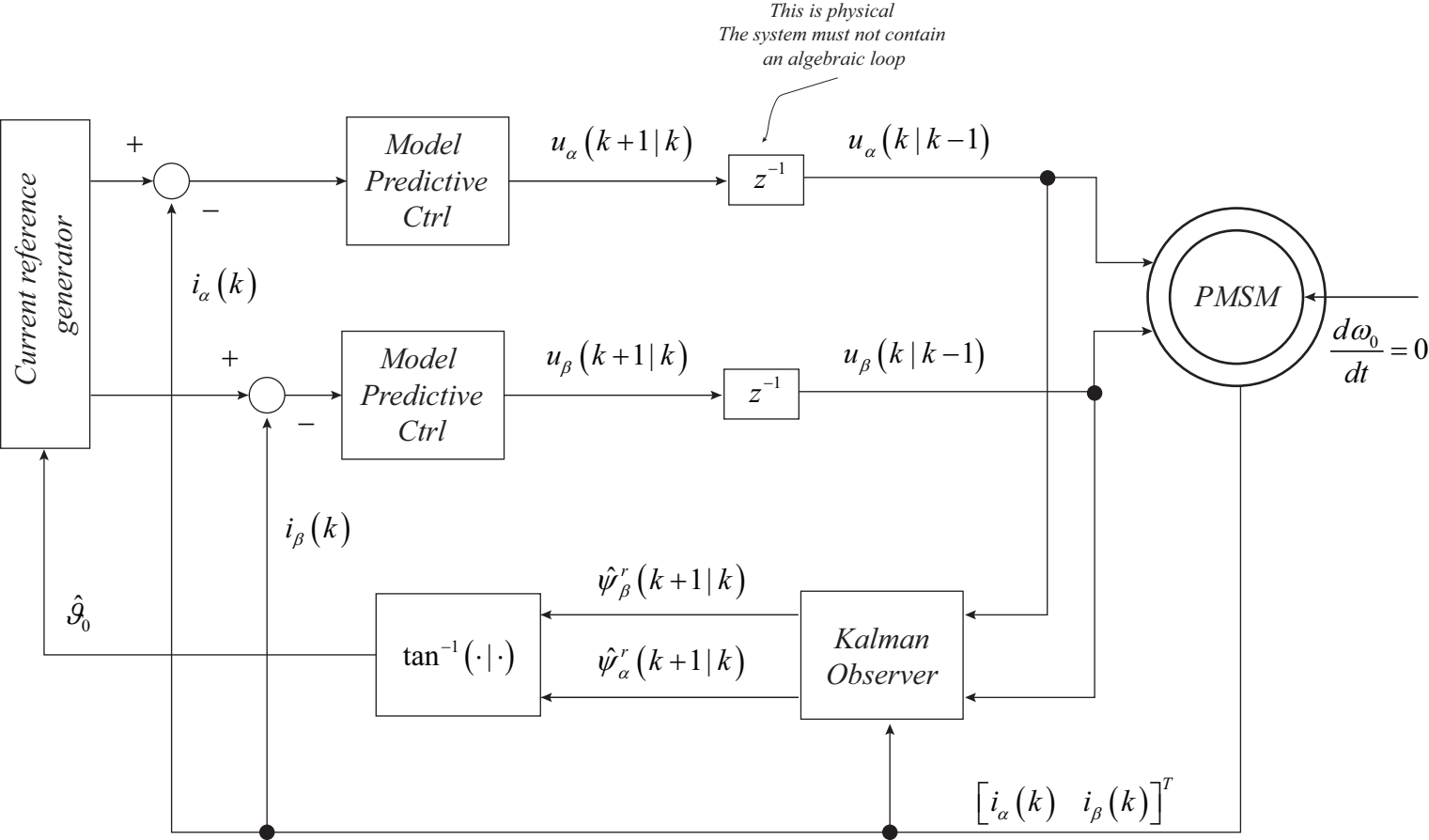
This is physical
The system must not contain
an algebraic loop
Current reference generator
+
−
Model Predictive Ctrl
$u_\alpha(k+1|k)$
z^{-1}
$u_\alpha(k|k-1)$
$i_\alpha(k)$
+
−
Model Predictive Ctrl
$u_\beta(k+1|k)$
z^{-1}
$u_\beta(k|k-1)$
PMSM
$\frac{d\omega_0}{dt}=0$
$i_\beta(k)$
$\hat{\vartheta}_0$
$\hat{\psi}_\beta^r(k+1|k)$
$\hat{\psi}_\alpha^r(k+1|k)$
$\tan^{-1}(\cdot|\cdot)$
Kalman Observer
$\left[i_\alpha(k) \quad i_\beta(k)\right]^T$